

The Cost of Exercising Rights: Consent Design in Data-Driven Platforms*

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Abstract

Digital platforms can grant users formal privacy rights while making those rights costly to exercise. This paper studies consent design in advertising-funded platforms that choose opt-out frictions before users choose platforms and consent. Higher refusal costs expand monetizable data and may improve data-driven quality, but burden users who refuse. With captive users, platform-chosen frictions are excessive when the marginal refusal burden exceeds the quality value of induced consent, and insufficient when the reverse holds. Under competition, privacy-sensitive users sort across platforms, so refusal costs can repel or attract demand. The welfare effects of friction caps also depend on business stealing.

Keywords: digital platforms, consent design, privacy, data-driven quality, platform competition

JEL Codes: D18, D61, K21, K24, L13, L51, L86

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1 Introduction

Digital regulation often gives users formal rights: to refuse data use, to avoid cross-service combination, to revoke consent, or to receive a non-personalized alternative. The effect of these rights depends on how they are implemented. A right to refuse data use can be easy to exercise, buried in menus, followed by repeated prompts, paired with degraded functionality, or offered only through a paid alternative.¹ A platform can therefore comply with a right while making its exercise costly.

The paper studies this mechanism in consent design. Consent is a useful setting because it supplies inputs to advertising and product quality. It expands the set of users whose data can be monetized through advertising and, in many services, improves search, recommendations, prediction, fraud detection, and matching quality (Bergemann and Bonatti, 2015; Wernerfelt et al., 2024; Hagiu and Wright, 2023; Schaefer and Sapi, 2023; Ullrich et al., 2024; Klein et al., 2025; Carballa-Smichowski et al., 2025). The same interface choice also determines which users bear refusal costs and which consent decisions change at the margin.

The platform’s choice is not the legal right alone, and it is not the amount of data collected conditional on consent. It is the cost of refusal. That cost changes who consents, how much usable data the platform has, how good the product becomes, and how demand responds. The paper asks when a platform wants to make refusal harder, when that choice is socially excessive in the captive-user benchmark, and how competition changes both demand and the welfare effect of friction caps.

The analysis endogenizes the cost of refusal and characterizes the resulting wedge. Existing work studies privacy levels and data collection (Casadesus-Masanell and Hervas-Drane, 2015; Dimakopoulos and Sudaric, 2018; Jakhu and Roy Chowdhury, 2025), opt-in and opt-out rights (Taylor and Wagman, 2008; Conti and Reverberi, 2021), personal data management or versioning (Cong and Matsushima, 2026; Choe et al., 2025; Bisceglia et al., 2025), and privacy choices when exercising privacy protection is costly (Montes et al., 2019; Abrardi et al., 2024). The strategic object here is different: the platform chooses the practical cost of exercising the right to refuse. Once that cost is endogenous, consent

¹Regulators and policy bodies treat these design margins as substantive rather than merely procedural. The European Commission’s Meta proceedings under the Digital Markets Act concern Article 5(2)’s consent requirement for cross-service data combination and the Commission’s view that users who refuse such consent must have access to a less personalized but equivalent alternative (European Commission, 2024c, 2025b). The Commission’s Apple proceedings concern separate DMA user-choice obligations and ask whether legally available choices are made burdensome or confusing in practice (European Commission, 2024a, 2025a). Separately, consumer-protection authorities raised concerns that Meta’s “pay or consent” interface pressured users to choose quickly (European Commission, 2024b). More generally, OECD (2022) documents dark commercial patterns that impair consumer choice in digital environments.

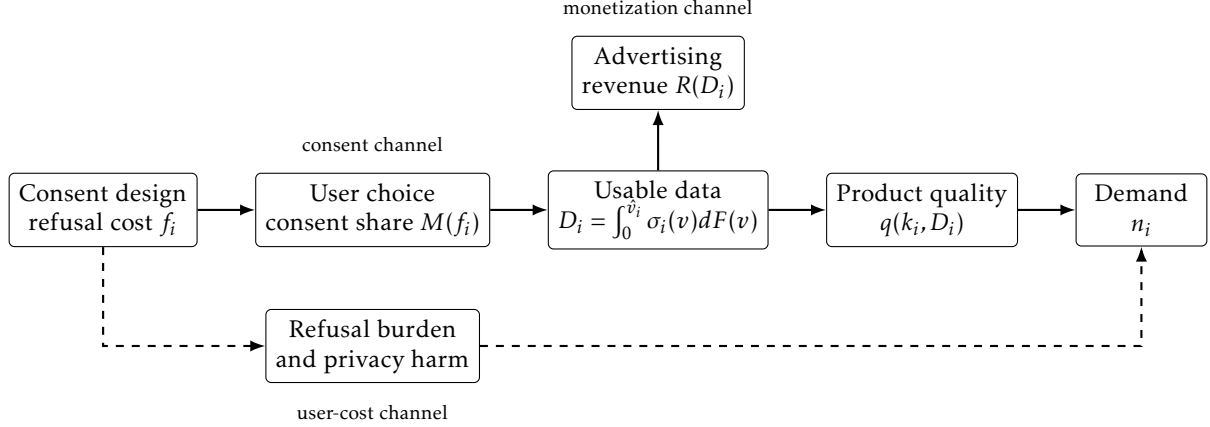


Figure 1: Strategic consent design. A higher refusal cost expands the consenting data pool, increasing monetization and data-driven quality. The same design makes refusal costly for users who do not consent. In competition, the same margin also changes demand through type-specific sorting.

design becomes a platform-design problem rather than a fixed regulatory constraint. The benchmark wedge can point in either direction when consent improves product quality, and competition can either reduce or amplify the incentive to raise refusal costs.

The model has two advertising-funded platforms. Consumers are horizontally differentiated and have heterogeneous privacy sensitivities. A consenting user may receive a personalization benefit but bears privacy disutility. A refusing user avoids that privacy cost but pays an opt-out friction chosen by the platform. A platform's usable data stock is the mass of its own users who consent, so platform choice and consent must be tracked jointly when privacy-sensitive users sort across platforms. Usable data increase advertising revenue and may improve service quality for all users on the platform. Platforms choose frictions; consumers then choose platforms and consent.

Figure 1 summarizes the mechanism. The first result characterizes the captive-user wedge between privately and socially optimal frictions. A platform raises refusal costs because doing so expands monetizable data. It does not internalize the first-order burden imposed on users who continue to refuse. Users who switch consent status are accounted for through the induced-data term and, in the baseline choice model, are locally indifferent at the consent threshold. Let $M(f)$ denote the consent share, D_f the induced change in usable data, and $q_D(k, D)$ the marginal quality value of data. Private friction is excessive when $1 - M(f) > q_D(k, D)D_f$ and insufficient when the inequality is reversed. The second case is driven by data-improved product quality: when $q_D = 0$, the captive-user benchmark implies locally excessive friction at every interior private optimum.

The second result is the competitive force. Refusal costs affect both the composition of demand and data-driven quality, so competition need not move design toward lower

frictions. In a symmetric equilibrium, a marginal increase in one platform’s refusal cost attracts demand only when

$$q_D(k, D) > \frac{2t[1 - M(f^*)]}{tM'(f^*) + 2M(f^*)[1 - M(f^*)]}.$$

Below this cutoff, higher refusal costs repel users. Above it, induced data improve quality enough to attract users despite the direct burden on those who refuse. The equilibrium effect of stronger competition also depends on the monetization value of the induced demand and data responses. The result links platform design to evidence that defaults, inattention, and switching frictions materially affect user choices in search and digital services (Allcott et al., 2025; Decarolis et al., 2025; Pape and Rossi, 2026).

The results yield testable policy objects. In the captive-user benchmark, a friction cap is justified when the refusal-cost harm exceeds the quality value of induced consent. In competition, the cap must also be evaluated against the business-stealing terms generated by platform choice. Service-equivalence rules and direct data-use caps concern related legal margins, but their welfare effects require primitives for service degradation, data scope, and external privacy harm (Acquisti et al., 2016; Goldfarb and Que, 2023; Choi et al., 2019; Rhodes and Zhou, 2024; Bisceglia et al., 2025). These margins are therefore kept outside the formal friction-cap result.

Regulatory Motivation

Two recent enforcement areas illustrate the model’s margins. In the United States, the search remedies in *United States v. Google* target distribution defaults, exclusive agreements, data access, and syndication rather than data scale alone. On September 2, 2025, the Department of Justice announced that the District Court prohibited Google from maintaining exclusive contracts relating to Search, Chrome, Google Assistant, and Gemini, required Google to make certain search index and user-interaction data available to rivals, and did not order a Chrome divestiture (United States Department of Justice, 2025a).² Although not a consent-design case, the remedies concern access points, defaults, and data-mediated quality, not data scale alone.

In the European Union, Article 5(2) of the Digital Markets Act restricts gatekeepers from processing, combining, or cross-using personal data across covered services without user consent (European Parliament and Council of the European Union, 2022). Recitals

²The court’s remedial package followed the August 2024 liability decision in which Google was found to have unlawfully maintained monopolies in general search and search text advertising. See the DOJ case page for the docket and related filings (United States Department of Justice, 2025b).

36 and 37 describe the associated policy concern: users should be able to refuse such processing through a less personalized but equivalent alternative, and that alternative should not be different or degraded except where degradation follows directly from the absence of the relevant data. The Commission’s Meta proceedings ask whether that consent choice is effective in practice; its Apple proceedings raise a parallel implementation concern under separate DMA user-choice obligations, namely whether legally available choices are made burdensome or confusing in practice (European Commission, 2024c, 2025b, 2024a, 2025a). This maps into the distinction between a formal right to opt out and the endogenous friction attached to exercising it.

The rest of the paper is organized as follows. Section 2 discusses the related literature on privacy, data-driven quality, platform competition, and digital choice architecture. Section 3 introduces the model. Section 4 derives the monopoly wedge and gives a parametric illustration. Section 5 studies platform competition with type-specific sorting. Section 6 derives the welfare effect of friction caps under competition. Section 7 states implications for platform design. Section 8 concludes. Proofs and extensions are in the Appendix.

2 Related Literature

This paper is closest to the literature on privacy and platform competition. Acquisti et al. (2016) and Goldfarb and Que (2023) provide broad accounts of privacy economics and its link to competition and advertising. The papers most directly related to the mechanism here make privacy or data collection a strategic choice. Casadesus-Masanell and Hervas-Drane (2015) show that competition need not lead firms to offer more privacy. Dimakopoulos and Sudaric (2018) and Jakhu and Roy Chowdhury (2025) let platforms choose how much data to collect and show that equilibrium data provision can be too high or too low. A related ambiguity arises here when induced consent improves product quality, but the choice variable is different. The firm does not choose a privacy level or a data quantity directly. It chooses how costly refusal is.

That distinction separates the paper from work in which consent rights or opt-out costs are imposed on the market. In Taylor and Wagman (2008), consumers can opt out of identification at an exogenous cost. In Conti and Reverberi (2021), opt-in privacy regulation interacts with quality when information and product quality are complementary. Cong and Matsushima (2026) study personal data management rights, including erasure, opt-out, portability, and ownership. In those papers, the legal or contractual margin is the object being evaluated. Here the right to refuse is already present. The platform chooses

the practical cost of exercising it.

Several related papers also study privacy choices when exercising privacy protection is costly. [Montes et al. \(2019\)](#) analyze competition when consumers can prevent the use of personal information by paying a privacy cost, and show that competition can reverse the monopoly effect of a higher privacy cost. [Abrardi et al. \(2024\)](#) study consent policy with time-inconsistent users and show that welfare effects depend on whether the design facilitates refusal. [Farronato et al. \(2025\)](#) provide field and structural evidence that consent-interface architecture affects cookie choices and consumer welfare. These papers motivate the refusal-cost margin studied here. The distinction is that the present model treats the cost of refusal as the platform’s strategic design variable and links it to consent, advertising revenue, data-driven quality, demand, and the welfare effect of friction caps.

The model needs data to matter for both monetization and product value. The advertising channel follows the standard two-sided-market and information-in-advertising logic in [Rochet and Tirole \(2003\)](#), [Armstrong \(2006\)](#), [Bergemann and Bonatti \(2015\)](#), and [Bergemann and Bonatti \(2024\)](#). The quality channel follows work on data-enabled learning. [Hagiu and Wright \(2023\)](#) distinguish across-user and within-user learning, and empirical studies document that data can improve search, prediction, targeting, and product quality, often with diminishing returns ([Schaefer and Sapi, 2023](#); [Ullrich et al., 2024](#); [Klein et al., 2025](#); [Carballa-Smichowski et al., 2025](#); [Wernerfelt et al., 2024](#)). These channels matter here because a refusal cost changes the data input into quality as well as the allocation of privacy costs.

The paper also relates to work on digital choice architecture. Defaults, beliefs, and inattention shape user behavior in search markets ([Allcott et al., 2025](#); [Decarolis et al., 2025](#)), and digital regulation often changes choices through interface design and compliance margins rather than through prices alone ([Pape and Rossi, 2026](#); [Kesler et al., 2025](#)). Policy and empirical work on dark patterns and consent interfaces documents that firms can alter the effective cost and salience of refusal ([OECD, 2022](#); [Sylwestrzak, 2026](#); [Zac et al., 2026](#)). That margin is used as the strategic instrument, yielding conditions for the competitive demand response and for the welfare effect of a friction cap.

Several adjacent papers study margins that are deliberately simplified in the baseline. [Fainmesser et al. \(2023\)](#) study how a digital business chooses data collection and data protection when data improve service value but expose users to privacy harms. Versioning and paid privacy options are analyzed by [Choe et al. \(2025\)](#) and [Bisceglia et al. \(2025\)](#). Information externalities in consent decisions are central in [Choi et al. \(2019\)](#). These contributions motivate the robustness discussion in the Appendix: prices, data-use intensity, externalities, and multidimensional data change the terms in the welfare comparison but

do not eliminate the basic consent-design channel.

3 Model

There are two platforms, $i \in \{1, 2\}$, located at the endpoints of a Hotelling line. Consumers are indexed by a location $x \in [0, 1]$ and a privacy sensitivity $v \in [0, \bar{v}]$. The distribution of v has cumulative distribution function F , density $g > 0$ on its support, and finite mean. Platforms charge zero monetary prices to users and earn revenue from advertising.

Each platform chooses an opt-out friction $f_i \geq 0$. The friction is the effective utility cost borne by a user who refuses personalized data use. It captures interface complexity, repeated prompts, loss of convenience, and other design choices that make refusal costly without modeling a separate transfer price.³ Choosing high frictions may also involve an implementation, legal, or reputational cost $K(f_i)$, where $K(0) = 0$, $K' \geq 0$, and $K'' \geq 0$. The case $K \equiv 0$ corresponds to frictions that are privately cheap to impose.

3.1 Consumers and Consent

If a consumer joins platform i , she chooses whether to consent to personalized data use. Let $a_i \in \{0, 1\}$ denote consent. A consenting user obtains a personalization benefit $b \geq 0$ and bears privacy cost ve , where $e > 0$ is the intensity of data use. A refusing user avoids the privacy cost but pays the opt-out friction f_i . For a given platform quality Q_i , the consumer's utility is

$$U_i^A(x, v) = u_0 + Q_i + b - ve - t|x - \ell_i|, \quad (1)$$

$$U_i^O(x, v) = u_0 + Q_i - f_i - t|x - \ell_i|, \quad (2)$$

where the superscripts A and O denote acceptance of personalized data use and opt-out, respectively, $\ell_1 = 0$, $\ell_2 = 1$, and $t > 0$ measures horizontal differentiation.

The consent decision is characterized by a threshold:

$$a_i(v) = 1 \iff v \leq \hat{v}(f_i) \equiv \frac{b + f_i}{e}. \quad (3)$$

³The formal model treats f_i as a reduced-form refusal cost. Section 6 derives the welfare effect of caps on this margin in the competitive model; Appendix C explains what must be added to evaluate service degradation, paid alternatives, or direct limits on data use. The model does not require platforms to have unconstrained control over refusal costs. The choice of f_i should be read as the residual interface or product-design margin left to the platform within a given legal and technological environment. Regulation can cap this margin or redirect pressure to other margins, such as service degradation, pricing, or data-use intensity.

Let

$$M(f_i) = F(\hat{v}(f_i)) \quad (4)$$

denote the population share that would consent on a platform with friction f_i . For interior thresholds,

$$M'(f_i) = \frac{g(\hat{v}(f_i))}{e} > 0. \quad (5)$$

Define the non-location payoff from the consent decision as

$$m_i(v; f_i) = \max\{b - ve, -f_i\}. \quad (6)$$

For a captive user base, the corresponding average non-location utility generated by platform i is

$$s_i = Q_i + \int_0^{\hat{v}(f_i)} (b - ve) dF(v) - f_i[1 - M(f_i)]. \quad (7)$$

By the envelope theorem,

$$\frac{\partial s_i}{\partial f_i} = \frac{\partial Q_i}{\partial f_i} - [1 - M(f_i)]. \quad (8)$$

Increasing f_i has two opposing effects on user value. It raises the consenting mass, which can increase data-driven quality, but it also makes refusal costly for users who still opt out.

3.2 Data, Quality, and Revenue

Let $\sigma_i(v)$ denote the share of type- v consumers who join platform i . Platform i 's usable data stock is the mass of its users who consent:

$$D_i = \int_0^{\hat{v}(f_i)} \sigma_i(v) dF(v). \quad (9)$$

Its total market share is $n_i = \int_0^{\bar{v}} \sigma_i(v) dF(v)$. In competition, consumers know v when choosing a platform, so platform membership and consent are generally sorted by privacy sensitivity. At a symmetric equilibrium with common friction, (9) gives $D_i = M(f)/2$. Data are productive in two ways. First, they raise the platform's advertising revenue per user or impression, summarized by a reduced-form function $R(D_i)$ with $R(0) \geq 0$, $R' > 0$, and $R'' \leq 0$. Second, they may improve service quality:

$$Q_i = q(k_i, D_i), \quad (10)$$

where $k_i \geq 0$ is quality capacity or data infrastructure, $q_D \geq 0$, $q_k > 0$, $q_{DD} \leq 0$, and $q_{kk} \leq 0$. The baseline analysis fixes $k_i = k$ and focuses on f_i ; Appendix B endogenizes k_i .

Profits are

$$\pi_i = n_i R(D_i) - C(k_i) - K(f_i), \quad (11)$$

where $C' > 0$ and $C'' > 0$. The reduced-form advertising revenue $R(D_i)$ replaces an explicit advertiser-pricing subgame. For example, usable data may determine targeting precision $\alpha(D_i)$, with $\alpha' > 0$ and $\alpha'' \leq 0$, and advertisers' willingness to pay for an impression may increase in that precision. If the platform sells impressions or targeted access on the advertising side, equilibrium advertising revenue per user can then be written as an increasing and weakly concave function of the usable data stock. The function $R(D_i)$ summarizes this data-monetization channel while leaving the advertiser market in the background.

The reduced-form objects M , R , and q keep the consent, monetization, and quality channels separate. The analysis uses only the following properties. The consent share is differentiable, with $M' > 0$ at an interior threshold. Advertising revenue is increasing in usable data, so $R' > 0$; weak concavity of R is used only for curvature and comparative-statics restrictions. Product quality is weakly increasing in usable data, so $q_D \geq 0$; the possibility that private friction is too low depends on this quality channel and disappears when $q_D = 0$. No result requires a functional form for M , R , or q outside the parametric illustration. Appendix D reports the threshold for a general distribution of privacy valuations and shows how behavioral consent, data-use externalities, multidimensional data, and dynamic learning change the welfare comparison.

3.3 Timing and Equilibrium

The timing is:

- (i) platforms choose opt-out frictions f_i ; Appendix B allows them also to choose quality capacity k_i ;
- (ii) consumers observe f_i , form equilibrium-consistent expectations about quality, and choose a platform and consent;
- (iii) advertising revenue is realized.

Because quality depends on the data stock induced by platform choice and consent, consumer choices are evaluated under rational expectations. Let Q_i^e denote expected

quality at the consumer-choice stage. Given expected qualities, a type- v consumer joins platform i rather than j on the share

$$\sigma_i(v) = \frac{1}{2} + \frac{Q_i^e - Q_j^e + m_i(v; f_i) - m_j(v; f_j)}{2t}, \quad i \neq j, \quad (12)$$

with the expression restricted to interior values. In equilibrium,

$$Q_i^e = Q_i^* = q(k_i, D_i^*), \quad D_i^* = \int_0^{\hat{v}(f_i)} \sigma_i^*(v) dF(v),$$

so users correctly anticipate the quality generated by the equilibrium data stock. The notation below suppresses the expectation superscript and writes utilities using equilibrium quality Q_i .

Assumption 1 (Strategy Set and Interior Region). *Strategies are restricted to a compact interval $[0, \bar{f}]$. The analysis focuses on equilibria at which consent thresholds are interior and the type-specific shares in (12) lie in $(0, 1)$ for all relevant types.*

Assumption 2 (Local Curvature). *At the equilibrium under study, each platform's marginal-profit condition is locally decreasing in its own friction. In the investment extension, the corresponding local second-order condition holds in (f_i, k_i) .*

The compact strategy set gives existence of a monopoly optimum, and of a best response for each platform, whenever the relevant continuation equilibrium is continuous in frictions. Interiority and local uniqueness are not imposed globally. Appendix A.3 gives primitive sufficient conditions: boundary marginal profits select an interior symmetric stationary point, and a sufficiently curved friction-cost function makes the relevant marginal-profit condition strictly decreasing. The competition results are local results around a symmetric interior equilibrium satisfying the data-quality stability condition stated in Proposition 2; they do not require global uniqueness of all equilibria.

4 Monopoly Benchmark

The monopoly benchmark isolates the central distortion. Set $n = 1$ and fix k . The platform chooses f to maximize

$$\pi(f) = R(D(f)) - K(f), \quad D(f) = M(f). \quad (13)$$

Omitting fixed capacity costs, total welfare is consumer surplus plus advertising surplus net of friction costs:

$$\begin{aligned}
W(f) = & \underbrace{q(k, D(f))}_{\text{quality}} + \underbrace{\int_0^{\hat{v}(f)} (b - ve) dF(v)}_{\text{consent surplus}} - \underbrace{f[1 - M(f)]}_{\text{opt-out cost}} \\
& + \underbrace{R(D(f))}_{\text{ad revenue}} - \underbrace{K(f)}_{\text{friction cost}} .
\end{aligned} \tag{14}$$

The benchmark welfare accounting treats advertising revenue $R(D)$ as advertising surplus net of advertiser-side costs. If advertising surplus differs from platform advertising revenue, the welfare condition acquires the additional term derived in Appendix D.

The platform's first-order condition is

$$R'(D)D_f = K'(f). \tag{15}$$

The social first-order condition is

$$\begin{aligned}
\underbrace{R'(D)D_f}_{\text{ad revenue}} + \underbrace{q_D(k, D)D_f}_{\text{quality gain}} - \underbrace{[1 - M(f)]}_{\text{opt-out harm}} = \underbrace{K'(f)}_{\text{friction cost}} .
\end{aligned} \tag{16}$$

The FOC differs from the private condition by two consumer-side terms: $[1 - M(f)]$ is the marginal utility loss borne by users who still refuse, while $q_D(k, D)D_f$ captures the marginal welfare gain from higher service quality made possible by additional user data. The monopolist internalizes advertising revenue but not these terms.

Proposition 1 (Private-Social Wedge). *At any interior friction level, define the marginal refusal-cost harm and the marginal quality value of induced consent as*

$$H_f(f) \equiv 1 - M(f), \quad L_f(f) \equiv q_D(k, D(f))D_f.$$

The wedge between the private and social marginal value of increasing opt-out friction is

$$\Delta(f) \equiv \frac{\partial \pi}{\partial f} - \frac{\partial W}{\partial f} = \underbrace{H_f(f)}_{\text{harm}} - \underbrace{L_f(f)}_{\text{quality gain}} . \tag{17}$$

At an interior private optimum, the privately chosen opt-out friction is locally excessive whenever

marginal refusal-cost harm exceeds the marginal quality gain:

$$H_f(f) > L_f(f), \quad (18)$$

and locally insufficient when the inequality is reversed.

Proof. See Appendix A.1. □

Proposition 1 gives the captive-user welfare comparison under this advertising-surplus convention. Private frictions can be excessive or insufficient. When $H_f(f) > L_f(f)$, the welfare loss borne by users who continue to refuse data use exceeds the quality gain generated by additional consent. Private frictions are then locally excessive. When $H_f(f) < L_f(f)$, the quality gain dominates the refusal-cost harm, and private frictions are locally insufficient. The second case requires data-driven quality. If $q_D(k, D) = 0$, then $L_f(f) = 0$ and private friction is locally excessive at every interior private optimum. Consequently, policies that uniformly reduce opt-out frictions need not improve welfare in the captive-user benchmark only when induced consent has enough quality value to offset the direct refusal-cost burden.

At an interior private optimum, the wedge gives a direct test for a marginal friction cap.

Corollary 1 (When Friction Reduction Is Robust). *At an interior private optimum, a marginal cap that lowers f raises welfare if and only if*

$$q_D(k, D(f)) < \bar{q}_D^M(f) \equiv \frac{1 - M(f)}{M'(f)}. \quad (19)$$

If $q_D(k, D(f)) > \bar{q}_D^M(f)$, the same marginal cap lowers welfare.

Proof. See Appendix A.2. □

The corollary identifies the threshold under which the simple policy intuition is valid. The threshold equates the quality gain from marginally induced consent, $q_D(k, D(f))M'(f)$, with the refusal-cost harm borne by users who continue to opt out, $1 - M(f)$. If the marginal quality value of data is below $\bar{q}_D^M(f)$, lowering opt-out frictions corrects coercion without sacrificing enough productive data to overturn the gain. If the marginal quality value lies above the threshold, the induced loss of consenting data dominates the reduction in refusal costs.

4.1 Parametric Illustration

To make the threshold transparent, suppose privacy sensitivities are uniformly distributed and the consent share is

$$M(f) = \frac{b+f}{e}, \quad f \in [0, e-b]. \quad (20)$$

Let data-driven quality be

$$q(k, D) = aD - \frac{\gamma}{2}D^2, \quad (21)$$

so that $q_D(k, D) = a - \gamma D$. In the monopoly benchmark, $D(f) = M(f)$ and $M'(f) = 1/e$. The friction-reduction condition in Corollary 1 becomes

$$1 - M(f) > (a - \gamma M(f))M'(f), \quad (22)$$

or equivalently,

$$e[1 - M(f)] > a - \gamma M(f). \quad (23)$$

This expression highlights the forces behind the welfare test. A friction cap is more likely to raise welfare when the remaining mass of non-consenting users is large, when marginal data have low quality value, or when quality exhibits strong diminishing returns. Conversely, when a is large or γ is small, the data induced by higher consent generate substantial quality value, and a friction cap can reduce welfare.

The same parameterization illustrates the competition demand threshold. In a symmetric duopoly, $D = M(f^*)/2$. Therefore $q_D(k, D) = a - \gamma M(f^*)/2$, and Proposition 2 implies that a marginal increase in one platform's refusal cost repels users when

$$a - \frac{\gamma}{2}M(f^*) < \frac{2t[1 - M(f^*)]}{tM'(f^*) + 2M(f^*)[1 - M(f^*)]}. \quad (24)$$

Since $M'(f^*) = 1/e$, this condition can be written as

$$a - \frac{\gamma}{2}M(f^*) < \frac{2te[1 - M(f^*)]}{t + 2eM(f^*)[1 - M(f^*)]}. \quad (25)$$

When the inequality is reversed, higher refusal costs attract users through the quality channel. The example therefore produces both demand regimes in closed form. If marginal data have little quality value, refusal costs mainly repel users. If marginal data strongly improve quality, the induced consenting data can raise service value enough to attract demand. The equilibrium response to stronger competition also depends on the monetization terms in Proposition 2's marginal-profit comparative static.

5 Platform Competition

Consider now a symmetric interior equilibrium of the two-platform game. Fix $k_i = k$, hold platform j 's friction fixed, and differentiate platform i 's payoff with respect to f_i . Let

$$M = M(f), \quad M' = M'(f), \quad D = \frac{M}{2},$$

and evaluate R , R' , and q_D at the symmetric data stock D . The platform's profit derivative is

$$\frac{\partial \pi_i}{\partial f_i} = \underbrace{n_i R'(D_i) \frac{\partial D_i}{\partial f_i}}_{\text{monetization}} + \underbrace{R(D_i) \frac{\partial n_i}{\partial f_i}}_{\text{demand}} - \underbrace{K'(f_i)}_{\text{friction cost}}. \quad (26)$$

The first term captures the marginal revenue value of usable data. The second term captures the demand effect of raising refusal costs. Since platform membership is sorted by privacy sensitivity, both demand and data responses are computed type by type.

Let

$$A \equiv \frac{d(Q_i - Q_j)}{df_i} \quad (27)$$

denote the quality gap opened by a marginal increase in platform i 's refusal friction. At the symmetric point, the type-specific platform share in (12) satisfies

$$\frac{\partial \sigma_i(v)}{\partial f_i} = \frac{A - \mathbf{1}\{v > \hat{v}(f)\}}{2t}. \quad (28)$$

For users who would consent, the direct payoff from platform i does not change when f_i increases. For users who would refuse, the direct payoff falls one-for-one. Hence the total demand response is

$$n_{if} = \int_0^{\bar{v}} \frac{\partial \sigma_i(v)}{\partial f_i} dF(v) = \frac{A - [1 - M]}{2t}. \quad (29)$$

The data response is different. Only users with $v \leq \hat{v}(f_i)$ contribute usable data, so

$$\begin{aligned} D_{if} &= \frac{M'}{2} + \int_0^{\hat{v}(f)} \frac{A}{2t} dF(v) \\ &= \frac{M'}{2} + \frac{MA}{2t}, \end{aligned} \quad (30)$$

$$D_{jf} = - \int_0^{\hat{v}(f)} \frac{A}{2t} dF(v) = - \frac{MA}{2t}. \quad (31)$$

The boundary term $M'/2$ in (30) is the mass of marginal users induced to consent on

platform i . Demand shifts enter the data stock only through the types that consent.

The sorting channel is first-order even without data-driven quality. If $q_D = 0$, then $A = 0$, so

$$n_{if} = -\frac{1-M}{2t}, \quad D_{if} = \frac{M'}{2}.$$

The platform loses only refusing users at the margin; its consenting mass rises by the newly induced consenters. For example, if v is uniform on $[0, 1]$, $e = 2$, $b = 0.2$, $t = 1$, and $f = 0.4$, then $M = 0.3$ and $M' = 0.5$. The demand response is $n_{if} = -0.35$, while the data response is $D_{if} = 0.25$.

The quality gap must be consistent with the induced data gap:

$$\begin{aligned} A &= q_D(D_{if} - D_{jf}) \\ &= q_D\left(\frac{M'}{2} + \frac{MA}{t}\right). \end{aligned} \quad (32)$$

Under the local stability condition

$$t > q_DM, \quad (33)$$

the fixed point has the unique local solution

$$A = \frac{q_DM'/2}{1 - q_DM/t}. \quad (34)$$

The condition in (33) is a slope condition on the data-quality feedback. It is not implied by interior Hotelling shares. If q_DM approaches t , the response of quality to the induced data gap becomes large, and the local comparative statics below are no longer well defined. The analysis therefore stays within the locally stable region.

Substituting (34) into (29) gives

$$n_{if} = \frac{q_DM'}{4(t - q_DM)} - \frac{1-M}{2t}. \quad (35)$$

Equivalently, higher refusal friction attracts demand if and only if

$$q_D > \bar{q}_D^C(f, t) \equiv \frac{2t[1 - M(f)]}{tM'(f) + 2M(f)[1 - M(f)]}. \quad (36)$$

Below this cutoff, friction repels users because the burden on refusing users dominates the induced quality gain. Above it, friction attracts users because the additional consenting data improve quality enough to offset the direct refusal burden.

Proposition 2 (Competition and Privacy Frictions). *At a symmetric interior equilibrium, suppose thresholds are interior, the type-specific Hotelling shares are interior, and the data-quality fixed point is locally stable: $t > q_D(k, D)M(f^*)$. Holding the rival's friction fixed, a marginal increase in f_i changes demand according to (35). Thus $n_{if} < 0$ if $q_D(k, D) < \bar{q}_D^C(f^*, t)$ and $n_{if} > 0$ if $q_D(k, D) > \bar{q}_D^C(f^*, t)$.*

If the symmetric marginal-profit condition is locally decreasing in f , the effect of a change in horizontal differentiation on the symmetric equilibrium friction is determined by

$$\Psi_t = \frac{R(D)[1 - M]}{2t^2} - \frac{q_D M'}{8(t - q_D M)^2} [2R(D) + MR'(D)]. \quad (37)$$

Lower t lowers the equilibrium friction when $\Psi_t > 0$ and raises it when $\Psi_t < 0$.

Proof. See Appendix A.4. □

Proposition 2 separates the demand regime from the equilibrium comparative static. The cutoff in (36) determines whether higher refusal costs repel or attract users. The equilibrium response to stronger competition also depends on the monetization terms in (37), because a change in t changes both the demand value and the data value of a marginal friction increase.

5.1 General Demand

The type-specific derivation does not depend on the linear Hotelling form. Let $\eta(v) > 0$ denote the local responsiveness of type- v demand to platform i 's vertical utility advantage at a symmetric point. Then

$$n_{if} = A \int_0^{\bar{v}} \eta(v) dF(v) - \int_{\hat{v}(f)}^{\bar{v}} \eta(v) dF(v), \quad (38)$$

$$D_{if} = \frac{M'}{2} + A \int_0^{\hat{v}(f)} \eta(v) dF(v), \quad (39)$$

$$D_{jf} = -A \int_0^{\hat{v}(f)} \eta(v) dF(v). \quad (40)$$

The quality-gap equation becomes

$$A = q_D \left[\frac{M'}{2} + 2A \int_0^{\hat{v}(f)} \eta(v) dF(v) \right]. \quad (41)$$

If

$$1 - 2q_D \int_0^{\hat{v}(f)} \eta(v) dF(v) > 0,$$

then

$$A = \frac{q_D M'/2}{1 - 2q_D \int_0^{\hat{v}(f)} \eta(v) dF(v)}. \quad (42)$$

The own-demand response is positive when the quality term in (38) exceeds the weighted mass of refusing users. Setting $\eta(v) = 1/(2t)$ gives (35) and (36).

5.2 Asymmetric Platforms

With asymmetric platforms, the same accounting applies locally after replacing the symmetric objects by platform-specific type shares and data responses. At a common friction and common friction cost, platform i has the larger marginal incentive to raise friction when

$$n_i R'_i(D_i) D_{if} + R_i(D_i) n_{if} > n_j R'_j(D_j) D_{jf} + R_j(D_j) n_{jf}, \quad (43)$$

where each derivative is computed from the type-specific demand and data-stock formulas above. This comparison is an off-equilibrium marginal-incentive comparison at a common friction. At each platform's own interior optimum, the same expression is pinned down by $K'_i(f_i)$ and therefore cannot by itself rank equilibrium frictions without assumptions on friction costs and curvature.

6 Welfare Under Competition

The monopoly benchmark identifies the captive-user wedge. In a competitive market, the welfare effect of refusal costs must also account for users who move across platforms, the rival's data stock, and the fact that advertising revenue attached to a marginal user may be transferred rather than created.

Let $\mathcal{X}_i(v)$ denote the set of locations of type- v consumers who choose platform i . Aggregate welfare is

$$W = \sum_i \int_0^{\bar{v}} \int_{\mathcal{X}_i(v)} [Q_i + m_i(v; f_i) - t|x - \ell_i|] dx dF(v) + \sum_i n_i R(D_i) - \sum_i K(f_i). \quad (44)$$

Advertising revenue is counted as surplus net of advertiser-side costs. Section D discusses how the welfare derivative changes when this convention is relaxed.

Consider a symmetric interior equilibrium and a marginal increase in f_i , holding f_j fixed. Let $M = M(f)$, $M' = M'(f)$, $D = M/2$, and $B = 1 - M$. Let A be the quality-gap response in (34). From Section 5,

$$n_{if} = \frac{A - B}{2t}, \quad D_{if} = \frac{M'}{2} + \frac{MA}{2t}, \quad D_{jf} = -\frac{MA}{2t}. \quad (45)$$

Hence

$$D_{if} + D_{jf} = \frac{M'}{2}. \quad (46)$$

A unilateral increase in f_i induces new consent on platform i and reallocates some existing consenting users across platforms. The reallocation affects private profits, but only the aggregate data response in (46) creates first-order aggregate data surplus at a symmetric point.

Differentiating (44) gives

$$W_{fi} = -\frac{B}{2} + \frac{q_D + R'}{2}(D_{if} + D_{jf}) - K'(f_i). \quad (47)$$

The first term is the direct cost imposed on users of platform i who refuse. The second term is the aggregate quality and advertising-surplus value of the induced data. The business-stealing component of demand cancels from welfare at the symmetric point because users leaving one platform join the other.

At an interior private equilibrium,

$$K'(f_i) = \frac{1}{2}R'D_{if} + Rn_{if}. \quad (48)$$

Combining (45)–(48) yields

$$W_{fi} = -\frac{B}{2} + \frac{q_DM'}{4} - \frac{R'MA}{4t} - \frac{R(A - B)}{2t}. \quad (49)$$

The last two terms capture the gap between private demand incentives and aggregate surplus. A platform values the data and revenue associated with users it attracts or loses, while aggregate welfare values only the surplus created net of transfers across platforms.

Proposition 3 (Friction Caps Under Competition). *At a symmetric interior private equilibrium satisfying the stability condition $t > q_DM(f^*)$, define A by (34), evaluated at f^* , and let*

$$\Gamma_f^C \equiv \frac{1 - M}{2} - \frac{q_DM'}{4} + \frac{R'MA}{4t} + \frac{R(A - [1 - M])}{2t}. \quad (50)$$

A marginal cap that lowers both platforms' refusal frictions raises welfare if and only if $\Gamma_f^C > 0$. It lowers welfare if and only if $\Gamma_f^C < 0$.

Proof. See Appendix A.5. □

Proposition 3 gives the welfare test for friction regulation in the symmetric competitive market. The sign of the demand response is not enough. When refusal costs repel users, platforms privately treat lost demand as lost revenue, even though part of that demand moves to the rival. When refusal costs attract users through data-driven quality, platforms value the gained demand in the opposite direction. A friction cap is welfare improving only when the direct reduction in refusal costs, net of the quality and advertising-surplus effects of induced data, dominates these equilibrium incentives.

6.1 Implications for Regulatory Instruments

The analysis separates three objects. The monopoly benchmark gives a captive-user welfare comparison. The competition analysis gives a demand response with type-specific sorting. Proposition 3 gives the welfare effect of a marginal friction cap in a symmetric competitive market.

Other instruments require their own primitives. A service-equivalence rule must specify how degradation of the non-personalized service enters user utility and data production. A data-use cap must specify whether the constrained use affects advertising revenue, quality, external privacy harm, or all three. These instruments can be compared to friction caps only after those margins are added to the same two-platform welfare function in (44).

7 Observable Implications for Platform Design

The results identify the empirical objects needed to evaluate refusal-cost regulation and the demand effects of consent design. The relevant objects are the consent response to refusal costs, $M'(f)$; the mass of users who still refuse, $1 - M(f)$; the induced change in usable data, D_f ; the marginal quality value of data, $q_D(k, D)$; the advertising value of data, $R'(D)$; advertising revenue per user, $R(D)$; and the demand response to opt-out frictions, n_f . The paper does not estimate these objects, but it identifies the comparisons that an experiment, regulatory change, or platform log analysis would have to recover.

Interface changes can identify the consent response to refusal costs. Examples include changes in reject-button salience, click paths, repeated prompts, default rules, or other

Table 1: Empirical objects

Model object	Empirical analogue
$M(f)$	Consent rate
$M'(f)$	Consent response to interface changes
D_f	Change in usable data after consent redesign
$q_D(k, D)$	Quality response to additional usable data
$R(D), R'(D)$	Advertising value of users and data
n_f	Demand response to opt-out frictions

design choices that alter the cost of refusal. Platform logs measure the resulting change in usable data, while product metrics such as search relevance, recommendation accuracy, fraud detection, retention, or task completion measure whether the induced data have a large quality effect.

The first implication concerns interface simplification in a captive-user environment. A policy that lowers f raises welfare in the captive-user benchmark in markets satisfying

$$1 - M(f) > q_D(k, D)D_f. \quad (51)$$

This is the empirical version of Corollary 1. The regime is more likely when refusal remains common despite existing frictions, when many users switch to refusal after reject options become salient, and when marginal consenting data have small effects on non-advertising quality.

The second implication concerns product quality. After a reduction in opt-out frictions, lower consent can reduce product quality. That quality loss overturns the direct refusal-cost gain in the captive-user benchmark only when

$$q_D(k, D)D_f > 1 - M(f). \quad (52)$$

Relevant quality measures depend on the service: search relevance, recommendation accuracy, matching quality, fraud detection, retention, or task completion. If these measures do not fall after consent becomes easier to refuse, the data-quality term is unlikely to be large enough to overturn the case for friction regulation.

The third implication concerns competition. In a symmetric market with privacy-sensitive sorting, the demand response to a unilateral increase in refusal costs is governed by

$$q_D(k, D) \geq \frac{2t[1 - M(f^*)]}{tM'(f^*) + 2M(f^*)[1 - M(f^*)]}. \quad (53)$$

Refusal costs repel users below this threshold and attract users above it. Stronger competitive pressure is therefore predicted to reduce symmetric equilibrium frictions only when the type-specific demand response and the marginal-profit comparative static both point toward lower frictions. Where data-driven quality is strong, higher refusal costs can attract demand.

The fourth implication concerns friction caps in a symmetric competitive market. Under the stability condition $t > q_D M$, the quality-gap response generated by a marginal increase in one platform's refusal friction is

$$A = \frac{q_D M' / 2}{1 - q_D M / t}.$$

Proposition 3 implies that a marginal cap raises welfare when

$$\frac{1 - M}{2} - \frac{q_D M'}{4} + \frac{R' M A}{4t} + \frac{R(A - [1 - M])}{2t} > 0. \quad (54)$$

The first two terms are the direct refusal-cost gain net of the aggregate quality value of induced consent. The last two terms adjust for competitive incentives: a platform values demand and data it gains from rivals, while welfare counts only aggregate surplus.

The fifth implication concerns heterogeneity across platforms. This is a comparison of marginal incentives at a common friction, not a comparison of the first-order condition evaluated at each platform's own optimum. Holding the consent technology and friction-cost schedule fixed, define

$$B_i(f) \equiv n_i R'_i(D_i) D_{if} + R_i(D_i) n_{if}, \quad (55)$$

with all objects evaluated at the same friction level and with the same friction-cost function K . A platform with larger $B_i(f)$ has a larger marginal benefit from raising friction at that common point. Under comparable local curvature, the implicit-function theorem then predicts a higher interior friction. The expression is large when marginal data are highly monetizable, when induced data strongly increase quality-based demand, or when refusal costs generate only limited demand losses at the common friction. At each platform's own interior optimum, $B_i(f_i^*) = K'_i(f_i^*)$ and the expression does not by itself rank equilibrium frictions.

The key empirical advantage is that these objects can be measured without recovering each user's privacy cost directly. Evidence on consent take-up identifies $M(f)$ and $M'(f)$; changes in the amount of usable data identify D_f ; and changes in service-quality metrics

identify whether $q_D(k, D)$ is large enough for induced consent to matter through product quality.

8 Conclusion

User rights do not operate only through their formal legal content. They also operate through the costs platforms attach to exercising them. In consent design, a platform can make refusal costly in order to expand the pool of consenting users, and that pool matters because it raises advertising revenue and may improve data-driven quality. Once this mechanism is modeled explicitly, the welfare effect of consent regulation is no longer pinned down by construction.

The benchmark result is that consent design creates a private-social wedge. In the captive-user benchmark, platforms set refusal costs too high when $1 - M(f) > q_D(k, D)D_f$ and too low when the inequality reverses. The second case requires data-driven quality; without it, friction is locally excessive. Competition adds a sorting channel: when refusal costs differ, platform membership varies with privacy sensitivity. The competitive data stock is therefore the mass of a platform's own consenting users. A marginal increase in refusal costs repels users when data-driven quality is weak, but can attract users when induced consent improves quality enough. This demand effect is not itself a welfare test. Under competition, the welfare effect of a friction cap also depends on business stealing and on how advertising revenue and data-driven quality enter aggregate surplus.

For platform designers and regulators, the same interface choice can either destroy user surplus or build productive data assets. Friction caps are appropriate when the welfare gain from reducing refusal costs exceeds the lost quality value of induced consent and the competitive incentives embedded in platform choice. Service-equivalence rules and direct limits on data use address different margins and require their own primitives.

The broader methodological point is that the desirability of consent regulation should not be built into the definition of the policy instrument. Once consent design is chosen by platforms, privacy regulation depends on equilibrium incentives, data productivity, and the source of the underlying distortion.

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A Proofs

A.1 Proof of Proposition 1

In the monopoly benchmark, $D(f) = M(f)$ and $D_f = M'(f) > 0$ for an interior threshold. Profit is

$$\pi(f) = R(D(f)) - K(f),$$

so

$$\frac{\partial \pi}{\partial f} = R'(D)D_f - K'(f).$$

Welfare is

$$W(f) = q(k, D(f)) + \int_0^{\hat{v}(f)} (b - ve) dF(v) - f[1 - M(f)] + R(D(f)) - K(f).$$

Differentiate the integral term. The upper-boundary term is

$$(b - e\hat{v})g(\hat{v})\hat{v}_f.$$

Using $\hat{v} = (b + f)/e$, this equals $-f g(\hat{v})\hat{v}_f$. Differentiating $-f[1 - M(f)]$ gives

$$-[1 - M(f)] + fM'(f).$$

The two boundary terms cancel because $M'(f) = g(\hat{v})\hat{v}_f$. Therefore

$$\frac{\partial W}{\partial f} = [R'(D) + q_D(k, D)]D_f - [1 - M(f)] - K'(f).$$

Subtracting the social derivative from the private derivative gives

$$\frac{\partial \pi}{\partial f} - \frac{\partial W}{\partial f} = [1 - M(f)] - q_D(k, D)D_f.$$

The local excessive and insufficient friction conditions follow immediately. □

A.2 Proof of Corollary 1

At an interior private optimum, the monopolist's first-order condition is

$$R'(D)D_f = K'(f).$$

From the welfare derivative in the proof of Proposition 1,

$$\frac{\partial W}{\partial f} = [R'(D) + q_D(k, D)]D_f - [1 - M(f)] - K'(f).$$

Substituting the private first-order condition gives

$$\frac{\partial W}{\partial f} = q_D(k, D)D_f - [1 - M(f)].$$

In the monopoly benchmark, $D(f) = M(f)$, so $D_f = M'(f) > 0$. Therefore a marginal cap that lowers f raises welfare if and only if

$$\frac{\partial W}{\partial f} < 0 \iff q_D(k, D(f)) < \frac{1 - M(f)}{M'(f)}.$$

□

A.3 Equilibrium and Curvature Conditions

This subsection records primitive sufficient conditions for the local equilibrium objects used in the text. The conditions are sufficient, not necessary.

In the monopoly benchmark, take the strategy set to be $[0, \bar{f}]$ and suppose M , R , and K are twice continuously differentiable on this interval, with $0 < \hat{v}(f) < \bar{v}$ for all $f \in [0, \bar{f}]$. The private marginal profit is

$$G^M(f) \equiv \frac{d\pi}{df} = R'(M(f))M'(f) - K'(f).$$

Continuity on the compact strategy set implies that a private optimum exists. If

$$G^M(0) > 0 \quad \text{and} \quad G^M(\bar{f}) < 0,$$

then neither boundary can solve the monopoly problem, and the intermediate value theorem gives at least one interior stationary point. A simple strict-concavity condition is

$$K''(f) > R''(M(f))[M'(f)]^2 + R'(M(f))M''(f) \quad \text{for all } f \in [0, \bar{f}]. \quad (56)$$

Under (56), G^M is strictly decreasing, so the interior stationary point is unique and is the unique monopoly optimum. With a uniform privacy distribution, $M''(f) = 0$; since $R'' \leq 0$, any strictly convex K is sufficient for strict concavity of monopoly profit. If K is flat

and R is linear, an interior optimum is not generic: the optimum is at a boundary unless the marginal advertising return is exactly constant at the cost slope. The friction-cost term therefore has an economic role. It captures implementation, legal, reputational, or operational costs that eventually offset the monetization gain from inducing more consent.

For the symmetric competitive analysis, define, for any candidate common friction f satisfying $t > q_D(k, M(f)/2)M(f)$,

$$A(f, t) = \frac{q_D(k, M(f)/2)M'(f)/2}{1 - q_D(k, M(f)/2)M(f)/t},$$

$$N(f, t) = \frac{A(f, t) - [1 - M(f)]}{2t}, \quad Z(f, t) = \frac{M'(f)}{2} + \frac{M(f)A(f, t)}{2t}.$$

Here $N(f, t)$ is the own-demand response n_{if} and $Z(f, t)$ is the own-data response D_{if} at a symmetric point. The symmetric marginal-profit condition is

$$G^C(f, t) \equiv \frac{1}{2}R'(M(f)/2)Z(f, t) + R(M(f)/2)N(f, t) - K'(f) = 0.$$

If $G^C(0, t) > 0$ and $G^C(\bar{f}, t) < 0$, then the symmetric marginal-profit condition has an interior solution. If, in addition,

$$K''(f) > \frac{\partial}{\partial f} \left[\frac{1}{2}R'(M(f)/2)Z(f, t) + R(M(f)/2)N(f, t) \right] \quad \text{for all } f \in [0, \bar{f}], \quad (57)$$

then $G^C(f, t)$ is strictly decreasing and the symmetric marginal-profit equation has a unique interior solution. If platform i 's payoff is concave in f_i in a neighborhood of f^* , holding $f_j = f^*$ fixed and using the locally stable continuation equilibrium, this solution is a local best response to itself. It is therefore a locally valid symmetric equilibrium candidate; uniqueness here is uniqueness of the symmetric marginal-profit solution, not global uniqueness of the two-dimensional best-response system. Condition (57) is the competitive analogue of (56): the curvature of the friction-cost function must dominate the curvature of the marginal revenue and demand benefits from raising friction along the symmetric path. The local comparative statics in Proposition 2 require only this local second-order condition at the symmetric equilibrium, $G_f^C(f^*, t) < 0$, together with the data-quality stability condition $t > q_D(k, D)M(f^*)$. They do not require global uniqueness of asymmetric equilibria.

A.4 Proof of Proposition 2

At a symmetric equilibrium, $f_i = f_j = f$, $n_i = n_j = 1/2$, and $D_i = D_j = D = M/2$. Let

$$A = \frac{d(Q_i - Q_j)}{df_i}$$

be the change in the quality gap caused by a marginal increase in f_i , holding f_j fixed. For type v ,

$$\sigma_i(v) = \frac{1}{2} + \frac{Q_i - Q_j + m_i(v; f_i) - m_j(v; f_j)}{2t}.$$

The derivative of $m_i(v; f_i)$ with respect to f_i is zero for $v < \hat{v}(f)$ and -1 for $v > \hat{v}(f)$. The kink at $v = \hat{v}(f)$ has measure zero for the demand derivative. Hence

$$\frac{\partial \sigma_i(v)}{\partial f_i} = \frac{A - \mathbf{1}\{v > \hat{v}(f)\}}{2t}.$$

Integrating over all types gives

$$n_{if} = \frac{A}{2t} - \frac{1 - M}{2t} = \frac{A - [1 - M]}{2t}.$$

For the data stock, the upper limit also moves:

$$D_i = \int_0^{\hat{v}(f_i)} \sigma_i(v) dF(v).$$

At the symmetric point, $\sigma_i(\hat{v}(f)) = 1/2$. Therefore

$$D_{if} = \sigma_i(\hat{v}(f))M' + \int_0^{\hat{v}(f)} \frac{A}{2t} dF(v) = \frac{M'}{2} + \frac{MA}{2t}.$$

The rival's consent threshold does not move, so

$$D_{jf} = \int_0^{\hat{v}(f)} \frac{\partial \sigma_j(v)}{\partial f_i} dF(v) = -\frac{MA}{2t}.$$

The quality gap must satisfy

$$A = q_D(D_{if} - D_{jf}) = q_D\left(\frac{M'}{2} + \frac{MA}{t}\right).$$

The local map from a conjectured quality-gap response to the realized response has slope $q_D M/t$ around the symmetric point. Thus $t > q_D M$ is the local stability and uniqueness condition for this data-quality feedback. If $t > q_D M$, this equation has the unique local solution

$$A = \frac{q_D M'/2}{1 - q_D M/t}.$$

Substitution into the demand derivative gives

$$n_{if} = \frac{q_D M'}{4(t - q_D M)} - \frac{1 - M}{2t}.$$

Since $t > q_D M$, $n_{if} > 0$ if and only if

$$\frac{q_D M'}{4(t - q_D M)} > \frac{1 - M}{2t}.$$

Multiplying by the positive denominator $4t(t - q_D M)$ yields

$$q_D M' t > 2(1 - M)(t - q_D M),$$

or

$$q_D \{tM' + 2M(1 - M)\} > 2t(1 - M).$$

Thus

$$n_{if} \geq 0 \iff q_D \geq \frac{2t(1 - M)}{tM' + 2M(1 - M)}.$$

It remains to compute the equilibrium comparative static. At the symmetric point, the marginal-profit condition is

$$\Psi(f, t) = \frac{1}{2}R'(D)D_{if} + R(D)n_{if} - K'(f) = 0.$$

Holding f fixed, M , M' , D , $R(D)$, $R'(D)$, and q_D do not vary with t . From

$$A = \frac{q_D M' t}{2(t - q_D M)},$$

the solved derivatives can be written as

$$n_{if} = \frac{q_D M'}{4(t - q_D M)} - \frac{1 - M}{2t}, \quad D_{if} = \frac{M'}{2} + \frac{q_D M M'}{4(t - q_D M)}.$$

Therefore

$$\frac{\partial n_{if}}{\partial t} = -\frac{q_D M'}{4(t - q_D M)^2} + \frac{1 - M}{2t^2}, \quad \frac{\partial D_{if}}{\partial t} = -\frac{q_D M M'}{4(t - q_D M)^2}.$$

Differentiating Ψ with respect to t gives

$$\Psi_t = \frac{1}{2} R'(D) \frac{\partial D_{if}}{\partial t} + R(D) \frac{\partial n_{if}}{\partial t}.$$

Substituting the preceding derivatives yields

$$\Psi_t = \frac{R(D)[1 - M]}{2t^2} - \frac{q_D M'}{8(t - q_D M)^2} [2R(D) + MR'(D)].$$

By the local second-order condition, $\Psi_f < 0$. The implicit-function theorem gives

$$\frac{df^*}{dt} = -\frac{\Psi_t}{\Psi_f},$$

so df^*/dt has the sign of Ψ_t . A decrease in t therefore lowers f^* when $\Psi_t > 0$ and raises f^* when $\Psi_t < 0$. $\square$

A.5 Proof of Proposition 3

At a symmetric interior equilibrium, let $f_i = f_j = f$, $n_i = n_j = 1/2$, $D_i = D_j = D = M/2$, and $B = 1 - M$. For each type v , let $x^*(v)$ be the location indifferent between the two platforms. Then $\mathcal{X}_i(v) = [0, x^*(v)]$ and $\mathcal{X}_j(v) = [x^*(v), 1]$. The consumer-surplus and transport part of welfare is

$$\mathcal{U} = \int_0^{\bar{v}} \left[\int_0^{x^*(v)} \{Q_i + m_i(v; f_i) - tx\} dx + \int_{x^*(v)}^1 \{Q_j + m_j(v; f_j) - t(1 - x)\} dx \right] dF(v).$$

Differentiating with respect to f_i , holding f_j fixed, gives no boundary term because the marginal consumer of each type is indifferent:

$$Q_i + m_i(v; f_i) - tx^*(v) = Q_j + m_j(v; f_j) - t[1 - x^*(v)].$$

The remaining derivative is the direct effect on platform i 's refusing users plus the quality effects on users of both platforms:

$$\mathcal{U}_{f_i} = - \int_{\hat{v}(f)}^{\bar{v}} \int_0^{1/2} dx dF(v) + n_i q_D D_{if} + n_j q_D D_{jf}.$$

At the symmetric point,

$$\mathcal{U}_{f_i} = -\frac{B}{2} + \frac{q_D}{2}(D_{if} + D_{jf}).$$

Advertising surplus is

$$\mathcal{A} = n_i R(D_i) + n_j R(D_j).$$

Its derivative is

$$\mathcal{A}_{f_i} = R(D)n_{if} + \frac{1}{2}R'D_{if} + R(D)n_{jf} + \frac{1}{2}R'D_{jf}.$$

Full coverage gives $n_{jf} = -n_{if}$, so

$$\mathcal{A}_{f_i} = \frac{R'}{2}(D_{if} + D_{jf}).$$

Thus

$$W_{f_i} = -\frac{B}{2} + \frac{q_D + R'}{2}(D_{if} + D_{jf}) - K'(f_i).$$

From Proposition 2,

$$D_{if} + D_{jf} = \left(\frac{M'}{2} + \frac{MA}{2t} \right) - \frac{MA}{2t} = \frac{M'}{2}.$$

Therefore

$$W_{f_i} = -\frac{B}{2} + \frac{(q_D + R')M'}{4} - K'(f_i).$$

At an interior private equilibrium,

$$K'(f_i) = \frac{1}{2}R'D_{if} + Rn_{if}.$$

Substituting

$$D_{if} = \frac{M'}{2} + \frac{MA}{2t}, \quad n_{if} = \frac{A - B}{2t}$$

yields

$$W_{f_i} = -\frac{B}{2} + \frac{(q_D + R')M'}{4} - \frac{R'}{2} \left(\frac{M'}{2} + \frac{MA}{2t} \right) - R \frac{A - B}{2t}.$$

The terms $R'M'/4$ cancel, leaving

$$W_{f_i} = -\frac{B}{2} + \frac{q_D M'}{4} - \frac{R' MA}{4t} - \frac{R(A - B)}{2t}.$$

By symmetry, the derivative of welfare with respect to a common change in both platforms' frictions is $2W_{f_i}$. A marginal cap lowers frictions, so it raises welfare if and only if $W_{f_i} < 0$. This condition is equivalent to $\Gamma_f^C = -W_{f_i} > 0$, with Γ_f^C defined in (50). $\square$

B Endogenous Quality Investment

Suppose platforms choose quality capacity before consumers choose platforms and consent. The variable k_i captures investments that make user data more productive, such as data infrastructure, model quality, engineering effort, or complementary assets. This extension identifies when a cap on opt-out friction also affects those investments.

Platform i chooses (k_i, f_i) to maximize

$$\pi_i = n_i R(D_i) - C(k_i) - K(f_i), \quad D_i = \int_0^{\hat{v}(f_i)} \sigma_i(v) dF(v), \quad Q_i = q(k_i, D_i). \quad (58)$$

The private marginal return to quality capacity is

$$\frac{\partial \pi_i}{\partial k_i} = n_i R'(D_i) D_{ik} + R(D_i) n_{ik} - C'(k_i). \quad (59)$$

Because k_i affects consumers through quality, demand responds through the same data-quality fixed point as in the competition analysis. At a symmetric point, let

$$A_k \equiv \frac{d(Q_i - Q_j)}{dk_i}.$$

Then $n_{ik} = A_k/(2t)$, $D_{ik} = MA_k/(2t)$, and $D_{jk} = -MA_k/(2t)$. Hence

$$A_k = q_k + q_D(D_{ik} - D_{jk}) = q_k + \frac{q_D MA_k}{t}.$$

Under $t > q_D M$,

$$n_{ik} = \frac{q_k}{2(t - q_D M)}. \quad (60)$$

Let

$$I_f \equiv \frac{\partial}{\partial f_i} [n_i R'(D_i) D_{ik} + R(D_i) n_{ik}] \quad (61)$$

be the effect of friction on the marginal private return to capacity, evaluated at an interior equilibrium. Under Assumption 2, the implicit-function theorem gives

$$\frac{dk_i^*}{df_i} = -\frac{I_f}{\partial^2 \pi_i / \partial k_i^2}.$$

Since $\partial^2 \pi_i / \partial k_i^2 < 0$, dk_i^* / df_i has the sign of I_f . Therefore a marginal cap that lowers f_i

lowers equilibrium investment if and only if

$$I_f > 0, \quad (62)$$

and raises equilibrium investment if and only if

$$I_f < 0. \quad (63)$$

The cross-effect can be decomposed as

$$I_f = \underbrace{S_i^R}_{\text{advertising return}} + \underbrace{S_i^D}_{\text{data productivity}} + \underbrace{N_i}_{\text{demand channel}}, \quad (64)$$

where

$$S_i^R \equiv [n_{if}R'(D_i) + n_iR''(D_i)D_{if}]D_{ik}, \quad (65)$$

$$S_i^D \equiv n_iR'(D_i)D_{ikf}, \quad (66)$$

$$N_i \equiv R'(D_i)D_{if}n_{ik} + R(D_i)n_{ikf}. \quad (67)$$

Thus the cap lowers investment if and only if

$$S_i^R + S_i^D > -N_i, \quad (68)$$

and raises investment if and only if $S_i^R + S_i^D < -N_i$. This extension gives no unconditional investment prediction. A friction cap reduces investment only when friction raises the marginal return to capacity. The investment effect is ambiguous because a lower friction changes both the amount of usable data and the demand response to quality investment.

C Additional Regulatory Margins

Service-equivalence rules and data-use limits act through margins not modeled as choices in the baseline game. A service-equivalence rule requires a primitive for the quality or utility loss imposed on users who refuse personalized data use. A data-use limit requires a primitive for the scope of permissible data processing and for any harm not internalized by the consenting user.

These instruments can be evaluated in the same welfare function as (44), but their derivatives depend on how the additional margin enters user utility, advertising revenue,

service quality, and external harm. For this reason, the formal welfare result in Section 6 is limited to marginal caps on refusal frictions. Extending the analysis to service degradation or data-use scope requires adding those variables to the platform-choice and consent equations before deriving the associated welfare test.

D Robustness and Boundary Cases

This section records robustness checks and additional margins that can be added to the welfare derivative.

D.1 Distribution of Privacy Valuations

The results do not require a uniform distribution of privacy sensitivities. The only distributional objects used in the thresholds are the consenting share

$$M(f) = F\left(\frac{b+f}{e}\right)$$

and its local responsiveness

$$M'(f) = \frac{g(\hat{v}(f))}{e}.$$

The monopoly cap threshold can therefore be written as

$$\bar{q}_D^M(f) = \frac{1 - M(f)}{M'(f)} = e \frac{1 - F(\hat{v}(f))}{g(\hat{v}(f))}. \quad (69)$$

The symmetric competition demand threshold is

$$\bar{q}_D^C(f^*) = \frac{2t[1 - M(f^*)]}{tM'(f^*) + 2M(f^*)[1 - M(f^*)]} = \frac{2te[1 - F(\hat{v}(f^*))]}{tg(\hat{v}(f^*)) + 2eF(\hat{v}(f^*))[1 - F(\hat{v}(f^*))]}. \quad (70)$$

Hence the relevant distributional objects are local: the mass of users who refuse, the density of users close to the consent margin, and the consenting mass that feeds back into quality-based demand. A distribution with many marginal users makes $M'(f)$ large and lowers the threshold at which data-driven quality overturns the usual friction-reduction logic. What matters is behavior at the consent margin, not the full shape of the distribution.

D.2 No Data-Quality Channel

The benchmark $q_D(k, D) = 0$ identifies the case in which the simple opt-out result is valid. In the monopoly case,

$$\Delta(f) = 1 - M(f) > 0$$

at every interior friction. Private friction is locally excessive, and a marginal cap that lowers f raises welfare. In the symmetric competition case,

$$q_D(k, D) = 0 < \frac{2t[1 - M(f^*)]}{tM'(f^*) + 2M(f^*)[1 - M(f^*)]},$$

so refusal costs repel users. Thus, when refusal costs attract demand or private friction can be too low, the mechanism is data-driven quality rather than the cost function for friction.

D.3 Behavioral Consent

The main model assumes that consumers understand the privacy cost ve and make welfare-relevant consent choices. If consumers underestimate privacy costs, consent no longer implements self-selection even when $f_i = 0$. Let perceived privacy cost be ρve with $\rho \in (0, 1]$, while welfare is evaluated using the true privacy cost ve . The perceived consent threshold is

$$\hat{v}_\rho(f_i) = \frac{b + f_i}{\rho e}.$$

Let

$$M_\rho(f_i) = F(\hat{v}_\rho(f_i)), \quad M'_\rho(f_i) = \frac{g(\hat{v}_\rho(f_i))}{\rho e}.$$

For a captive user base, $D_\rho(f) = M_\rho(f)$, so $D_{\rho f} = M'_\rho(f)$. True welfare is

$$W_\rho(f) = q(k, D_\rho(f)) + R(D_\rho(f)) - K(f) + \int_0^{\hat{v}_\rho(f)} (b - ve) dF(v) - f[1 - M_\rho(f)].$$

Differentiating the user-surplus terms gives

$$\frac{d}{df} \left[\int_0^{\hat{v}_\rho(f)} (b - ve) dF(v) - f[1 - M_\rho(f)] \right] = -[1 - M_\rho(f)] - (1 - \rho)e\hat{v}_\rho(f)M'_\rho(f).$$

The additional term is the true welfare loss from marginal users who consent only because they underweight privacy harm. At an interior private optimum, $K'(f) = R'(D_\rho)D_{\rho f}$, so

$$\frac{\partial W_\rho}{\partial f} = q_D(k, D_\rho)D_{\rho f} - [1 - M_\rho(f)] - (1 - \rho)e\hat{v}_\rho(f)M'_\rho(f).$$

A marginal cap that lowers f raises true welfare if and only if

$$1 - M_\rho(f) + (1 - \rho)e\hat{v}_\rho(f)M'_\rho(f) > q_D(k, D_\rho)D_{\rho f}.$$

For $\rho = 1$, the additional term vanishes and the benchmark condition is recovered. For $\rho < 1$, lowering refusal costs has an extra welfare benefit because it prevents some users from consenting on the basis of understated privacy costs.

D.4 Additional Margins

The following variants add terms to the welfare derivative. First, suppose the platform can charge a user price or offer a paid non-personalized version. Profit then includes user revenue, for example $n_i[p_i + R(D_i)] - K(f_i)$, and the welfare effect of lowering f_i must include the induced price response. Paid privacy options also create a screening margin studied in related models of versioning and privacy menus (Choe et al., 2025; Bisceglia et al., 2025).

Second, suppose data-use intensity is endogenous. If the platform chooses both f_i and e_i , consent satisfies $M(f_i, e_i) = F((b + f_i)/e_i)$. A cap on f_i then improves the refusal margin but may induce adjustment in e_i . This does not make friction regulation irrelevant; it shows why direct limits on data use are distinct from interface regulation.

Third, if data generate information externalities, welfare contains an additional harm term $X(D_i)$ that is not internalized by the consenting user. The derivative of this term must be added to the friction-cap condition using the same data responses as the main welfare analysis. Information externalities therefore change the value of induced consent and strengthen the role of data-use limits (Choi et al., 2019).

Finally, if data are multidimensional or dynamically accumulated, replace the scalar quality term with the relevant marginal value of the induced data bundle. With a data vector $\mathbf{D}_i$, the quality term is $\nabla_{\mathbf{D}}q(k_i, \mathbf{D}_i) \cdot \mathbf{D}_{if}$. With dynamic learning, the right-hand side becomes the discounted quality value of the future data stock induced by current consent. Policy implementation then depends on measuring the marginal value of the induced data bundle.

D.5 Advertising Surplus

The baseline welfare function counts advertising revenue as surplus net of advertiser-side costs. If advertising revenue differs from advertising surplus, replace $R(D_i)$ in welfare by an advertising-surplus function $S(D_i)$ while keeping $R(D_i)$ in platform profit. In the monopoly benchmark, welfare becomes

$$W(f) = q(k, D(f)) + \int_0^{\hat{v}(f)} (b - ve) dF(v) - f[1 - M(f)] + S(D(f)) - K(f).$$

At an interior private optimum, $K'(f) = R'(D)D_f$, so

$$\frac{\partial W}{\partial f} = q_D(k, D)D_f + [S'(D) - R'(D)]D_f - [1 - M(f)].$$

Thus a marginal cap that lowers f raises welfare if and only if

$$1 - M(f) > [q_D(k, D) + S'(D) - R'(D)]D_f.$$

The benchmark condition in Corollary 1 is the special case $S' = R'$.

In competition, replace $R(D_i)$ in (44) by $S(D_i)$ while keeping $R(D_i)$ in platform profit. The private first-order condition is unchanged, but the welfare derivative uses $S'(D)$ in place of $R'(D)$ in the aggregate advertising-surplus term. At a symmetric competitive equilibrium,

$$W_{f_i} = -\frac{1 - M}{2} + \frac{(q_D + S')M'}{4} - \left[\frac{1}{2}R'D_{if} + Rn_{if} \right].$$

Thus a wedge between advertising revenue and advertising surplus introduces an additional monetization term into the welfare comparison.